Social Network Analysis (SNA)

(B Tech III CSE)

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Week#1 – Jan 7, 2026

Background and build up – What? Why?

- In the modern world, power, influence, risk, and information rarely operate in isolation.
- They propagate through **networks**:
- Ideas spread through social ties
- Malware spreads through communication graphs
- Capital flows through financial networks
- Trust and misinformation move through online platforms
- Terror cells, fraud rings, and botnets organize relationally, not hierarchically

What? Why?

- SNA provides the analytical language and tools to **see, measure, and reason about the structures discussed in the first slide**.
- Classical analysis asks:
- Who is rich?
- Who is influential?
- Who is risky?
- SNA reframes the question:
- **Who is connected to whom?**
- **Who bridges otherwise disconnected groups?**
- **Who controls information flow?**
- **Who is structurally replaceable—and who is not?**

What?

- Two individuals with identical qualifications or resources may have radically different outcomes purely due to **network position**.
- This shift—from attributes to relationships—is the intellectual foundation of SNA.

My portion SNA

Any two of first three + Case Studies

1. Community Detection in Social Networks
2. Information Diffusion and Cascade Behaviour in Social Networks
3. Graph Representational Learning
4. Case Studies

Why? How?

- Networks are intuitive:
- Friendship circles in a classroom
- Professional referrals in careers
- Supply chains linking manufacturers and suppliers
- Online platforms connecting users, content, and advertisers
- Yet intuition fails at scale.

Why? How?

- Human cognition struggles to reason beyond a few dozen relationships.
- Modern networks involve:
- Millions to billions of nodes
- Dynamic, evolving connections
- Hidden structures (communities, hierarchies, influence paths)
- SNA exists to **formalize intuition** and make large-scale relational systems analytically tractable.

Example - Designing Digital Library Access

- Community, Population, and Classification (e.g. SVNIT)
- Groups – Faculty, Research Scholars, Masters, Senior UG, Junior UG, Staff
- Privileges and Priorities, Weight assignment
- RBAC Role Based Access Control
- Concurrent users limit
- Kicking out users – who have been using for long period of time
- Use CPU scheduling or Memory management techniques to select outgoing candidate

Yami Gautam Dhar (First week of Jan 2026)

- Blown whistle about paid reviews for the movie release in Bollywood
- Special Premiere for journalists and media houses and influencers
- Positive Reviews – would ensure opening 2 days +15% business
- If not paid, negative reviews and negative investigations do not allow film to take off….
- For 3rd day onwards, its quality and appeal which can take film forward
- Chinese delegation met Venezuelan President just an hour before his capture by US forces ……

Terminologies

- Control
- Fake news and Misinformation
- Trust management
- Subjective logic
- Ratings
- Reputation
- Recommendation

Case study#1
One Message. Injuries and Deaths. No Leader

- **The Incident**
- In a small town, a message begins circulating on **WhatsApp**:
- “Child kidnappers spotted in nearby villages. Stay alert. Share widely.”
- There is no author.
- No verified source.
- No central organizer.

Effect

- Within 48 hours:
- The message has been forwarded thousands of times
- Panic spreads across villages
- Vigilant groups form spontaneously
- Innocent travelers are attacked
- Several people are injured and few killed
- No single individual planned this outcome.
- No organization issued orders.
- Yet the violence was *coordinated in effect*.

The Puzzle (Why Traditional Thinking Fails)

- Let us take up few questions:
- Who caused this?
- Who should be held responsible?
- Where did control reside?
- Why did *this* message spread, but thousands of others die out?
- If we apply **traditional analysis**, we fail:
- There is no leader to arrest
- No headquarters to raid
- No single account with massive reach
- No central decision-maker

Why? - This was not a hierarchical system. It was a **network phenomenon**.

Reframing the Problem (Network Lens)

- **What actually mattered was not *who said it*, but:**
- Who forwarded to whom
- Which groups were densely connected
- Which individuals bridged villages
- Which nodes amplified fear fastest
- How trust flowed through family, caste, and locality ties
- **The violence emerged from:**
- Local clustering
- High trust edges
- Repeated reinforcement
- Absence of corrective ties
- This is not psychology alone. This is **network structure** producing real-world outcomes.

Takeaways

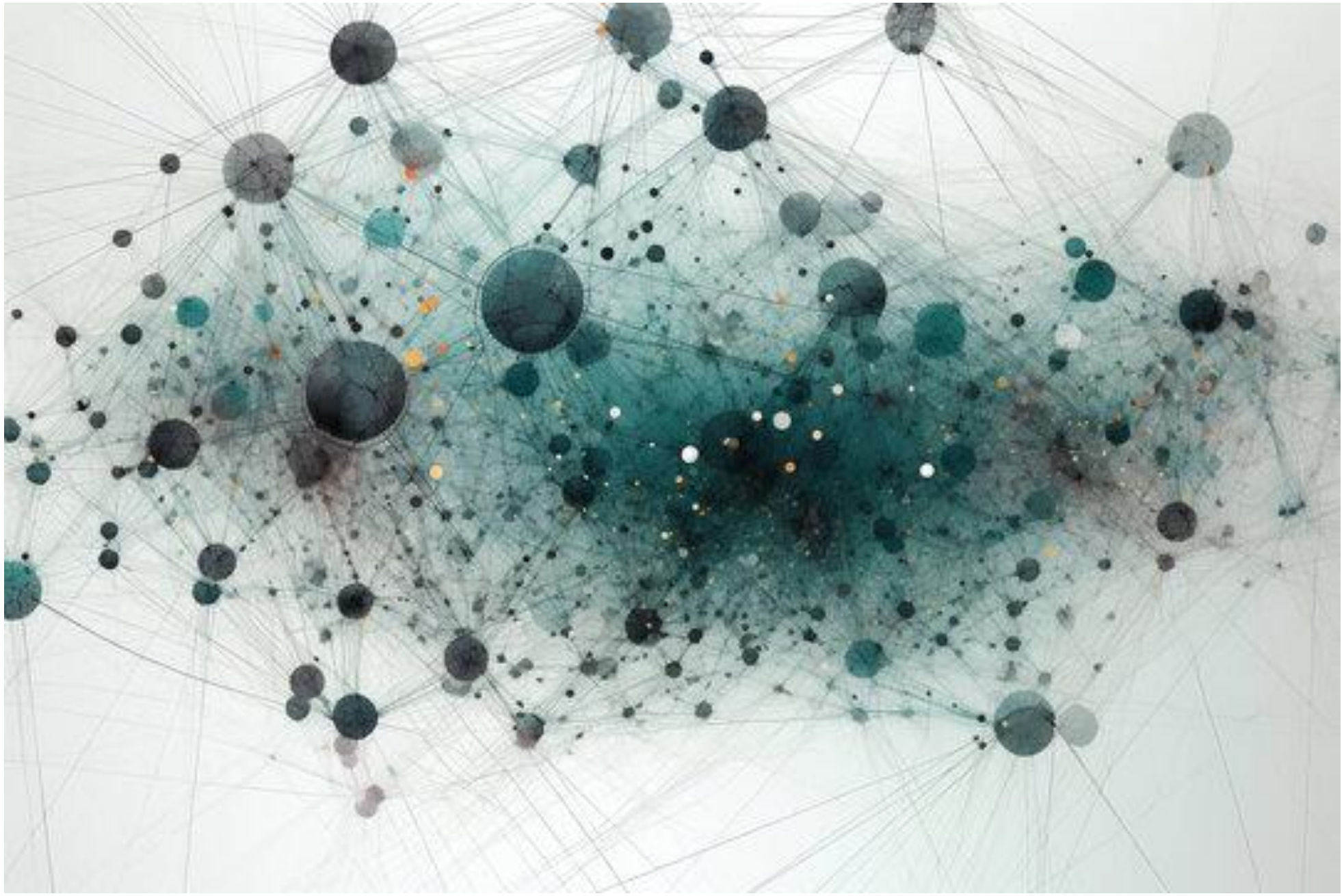
- No single message is dangerous in isolation.
- A message becomes dangerous when embedded in the wrong network structure.

Now escalate the stakes…

- If you were:
- A district administrator, A platform designer at **Meta**
- A law enforcement analyst, A national security planner
- **What would you do?**
- Ban the message? (Too late)
- Arrest senders? (Which ones?)
- Shut down the platform? (At what cost?)
- Issue clarifications? (Why don’t they propagate?)
- **Without understanding the network,**
- **every intervention is blunt, late, or counterproductive.**

Welcome to Social Network Analysis

- Nothing about this crisis makes sense if you look at individuals. Everything about it becomes intelligible when you look at the network
- SNA allows us to:
- Identify **super-spreaders without high follower counts**
- Detect **bridging nodes** between communities
- Understand **cascade thresholds**
- Predict where panic will jump next
- Design **surgical interventions** instead of blanket bans



What SNA Explains That Other Methods Miss

- **Traditional data analysis excels at**:
- Classification
- Prediction based on features
- Aggregated trends
- **SNA excels at:**
- Influence and contagion
- Collective behavior
- Emergent risk
- Coordination and collusion
- Systemic fragility